

Stony Brook University
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ASSIGNMENT 4

SOLUTIONS

INSTRUCTIONS: Show all your work to get full points. Please also cut and paste at the end of your submission the R code you have used for the last two problems.

1. Multiple Regression Model

The following table reports the results of three regressions of hourly earnings (*AHE*) on a number of regressors using a survey of full time workers in U.S. in 1998. The highest educational achievement for each worker was either a high school diploma or a bachelor's degree. Workers' ages ranged from 25 to 34 years. The data set also contained information on the region of the country where the person lived.

Regressor	(1)	(2)	(3)
College (X_1)	5.46 (0.21)	5.48 (0.21)	5.44 (0.21)
Female (X_2)	-2.64 (0.20)	-2.62 (0.20)	-2.62 (0.20)
Age (X_3)		0.29 (0.04)	0.29 (0.04)
Northeast (X_4)			0.69 (0.3)
Midwest (X_5)			0.60 (0.28)
South			-0.27 (0.26)
Intercept	12.69 (0.14)	4.40 (1.05)	3.75 (1.06)
Summary Statistics			
SER	6.27	6.22	6.21
R^2	0.176	0.190	0.194
$\bar{R}^2$			
n	4000	4000	4000

- *AHE* = average hourly earnings (in 1998 dollars)
- *College* = binary variable (1 if college, 0 if high school)
- *Female* = binary variable (1 if female, 0 if male)
- *Age* = age (in years)
- *Northeast* = binary variable (1 if Region = Northeast, 0 otherwise)
- *Midwest* = binary variable (1 if Region = Midwest, 0 otherwise)
- *South* = binary variable (1 if Region = South, 0 otherwise)
- *West* = binary variable (1 if Region = West, 0 otherwise)

- Compute $\bar{R}^2$ for every regression.
- Consider regression (1). Do workers with college degree earn more, on average, than workers with only high school degree? How much more?
- Consider regression (1). Do men earn more than women on average? How much more?
- Consider regression (2). Is Age an important determinant of earnings? Explain.
- Consider regression (2). John is a 31-year-old male without a college degree while Bob is a 24 year-old male with college degree. Predict John's and Bob's earnings.

- (f) Consider regression (3). Do there appear to be important regional differences? Why or why not?
- (g) Consider regression (3). Why was *West* excluded? What would happen if it was included?

Solution

- (a) $\bar{R}_{(1)}^2 = 1 - \frac{4000-1}{4000-2-1}(1 - R_{(1)}^2) = 1 - \frac{3999}{3997}(1 - 0.176) = 0.1756$, $\bar{R}_{(2)}^2 = 1 - \frac{4000-1}{4000-3-1}(1 - R_{(2)}^2) = 1 - \frac{3999}{3996}(1 - 0.190) = 0.1894$ and $\bar{R}_{(3)}^2 = 1 - \frac{4000-1}{4000-6-1}(1 - R_{(3)}^2) = 1 - \frac{3999}{3993}(1 - 0.194) = 0.1928$.
- (b) On average, a worker with a college degree earns 5.46/hour more for than workers without the degree.
- (c) On average, male workers earn 2.64/hour more than female workers.
- (d) Yes, age is an important determinant of earnings. The *t*-statistic is $0.29/0.04 = 7.25$, implying that the coefficient on age is statistically significant at the 1% level. In terms of the economic impact, note that 0.29/hour is equivalent to $0.29 \times 40 \times 4 \times 12 = \$556.8/\text{year}$. For each additional year the worker ages she gets 556.8 additional dollars on average.
- (e) John's expected earnings are $4.40 + 5.48 \times 0 - 2.62 \times 0 + 0.29 \times 31 = \13.39 per hour. Bob's expected earnings are $4.40 + 5.48 \times 1 - 2.62 \times 0 + 0.29 \times 24 = \16.84 per hour.
- (f) First, note that, except for South, the coefficients on the regional variables are statistically significant at $\alpha = 0.05$ significance level. On average, workers in the Northeast earn \$0.69 more per hour than workers in the West, workers in the Northeast earn \$0.60 more per hour than workers in the West, and workers in the South earn \$0.27 less than workers in the West, all controlling for other variables in the regression. Then, we can claim that there are important regional differences.
- (g) The regressor *West* is omitted to avoid perfect multicollinearity. If *West* is included, then the intercept can be written as a perfect linear function of the four regional regressors and the *OLS* method will fail. This is the so called *dummy variable trap* problem.

2. Data were collected from a random sample of 220 home sales from a community in 2013. Let *Price* denote the selling price in (in \$1000), *BDR* denote the number of bedrooms, *Bath* denote the number of bathrooms, *Hsize* denote the size of the house (in square feet), *Lsize* denote the lot size (in square feet), *Age* denote the age of the house (in years), and *Poor* denote a binary variable that is equal to 1 if the condition of the house is reported as "poor". An Estimated regression yields

$$\widehat{Price} = 119.2 + 0.485BDR + 23.4 Bath + 0.156 Hsize + 0.002 Lsize + 0.090 Age - 48.8 Poor, \bar{R}^2 = 0.72, SER = 41.5.$$

(23.9) (2.61) (8.94) (0.011) (0.00048)

(0.311) (10.5)

- (a) Suppose that a homeowner converts part of an existing family room in her house into a new bathroom. What is the expected increase in the value of the house? Suppose that a homeowner adds a new bathroom to her house, which increases the size of the house by 100 square feet. What is the expected increase in the value of the house?
- (b) What is the loss in value if a homeowner lets his house run down so that its condition becomes "poor"?
- (c) Compute the R^2 for the regression.
- (d) Is the coefficient on *BDR* statistically significantly different from zero?
- (e) Typically five-bedroom houses sell for much more than two-bedroom houses. Is this consistent with your answer to (d) and with the regression more generally?

- (f) A homeowner purchases 2000 square feet from an adjacent lot. Construct a 99% confidence interval for the change in the value of her house.
- (g) Lot size is measured in square feet. Do you think that another scale might be more appropriate? Why or why not?
- (h) The F -statistic for omitting BDR and Age from the regression is $F = 0.08$. Are the coefficients on BDR and Age statistically different from zero at the 10% level?

Solution

- (a) If the homeowner converts the room into a new bathroom, then the expected increase in the price is: \$23,400 (recall that Price is measured in \$1000s). By adding a bathroom which increases the size by 100 square feet we have that $\Delta BDR = 1$, $\Delta Hsize = 100$. The resulting expected change in price is $23.4 + 0.156 \times 100 = 39.0$ thousand dollars or \$39,000.
- (b) The loss is \$48,800.
- (c) $R^2 = 1 - \frac{n-k-1}{n-1}(1 - \bar{R}^2) = 0.727$.
- (d) The t -statistic is $\frac{0.485}{2.61} = 0.186 < 1.96$. Therefore, the coefficient on is not statistically significantly different from zero.
- (e) The coefficient on BDR measures the partial effect of the number of bedrooms holding house size ($Hsize$) constant. Yet, the typical 5-bedroom house is much larger than the typical 2-bedroom house. Thus, the results in (a) says little about the conventional wisdom.
- (f) The 99% confidence interval for effect of lot size on price is $2000 \times [.002 \pm 2.58 \times .00048]$ or 1.52 to 6.48 (in thousands of dollars).
- (g) Choosing the scale of the variables should be done to make the regression results easy to read and to interpret. If the lot size were measured in thousands of square feet, the estimate coefficient would be 2 instead of 0.002.
- (h) The 10% critical value from the $F_{2,\infty}$ distribution is 2.30. Because $0.08 < 2.30$, the coefficients are not jointly significant at the 10% level.

3. In a study relating college grade point average to time spent in various activities, you distribute a survey to several students. The students are asked how many hours they spend each week in four activities: studying, sleeping, working, and leisure. Any activity is put into one of the four categories, so that for each student, the sum of hours in the four activities must be 168.

- (a) In the model

$$GPA = \beta_0 + \beta_1 study + \beta_2 sleep + \beta_3 work + \beta_4 leisure + u,$$

does it make sense to hold *sleep*, *work* and *leisure* fixed, while changing *study*?

- (b) Explain why this model violates the *no perfect multicollinearity* assumption.
- (c) How could you reformulate the model so that its parameters have a useful interpretation and it satisfies the *no perfect multicollinearity* assumption?

Solution

- (a) No. By definition, $study + sleep + work + leisure = 168$. Therefore, if we change study, we must change at least one of the other categories so that the sum is still 168.
- (b) From part (a), we can write, say, *study* as a perfect linear function of the other independent variables: $study = 168 - sleep - work - leisure$.

- (c) Simply drop one of the independent variables, say *leisure*:

$$GPA = \beta_0 + \beta_1 \text{study} + \beta_2 \text{sleep} + \beta_3 \text{work} + u$$

4. Consider the multiple regression model containing three independent variables, under the four assumptions discussed in class:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + u.$$

You are interested in estimating the sum of the parameters on x_1 and x_2 ; call this $\theta_1 = \beta_1 + \beta_2$.

- Show that $\hat{\theta}_1 = \hat{\beta}_1 + \hat{\beta}_2$ is an unbiased estimator of θ_1 .
- Find $\text{Var}(\hat{\theta}_1)$ in terms of $\text{Var}(\hat{\beta}_1)$, $\text{Var}(\hat{\beta}_2)$, and $\text{Cov}(\hat{\beta}_1, \hat{\beta}_2)$. *Hint: use the formula for the variance of the sum of two random variables.*
- The expression for $\text{Var}(\hat{\theta}_1)$ that you found in the previous item is the *theoretical variance* of $\hat{\theta}_1$. How would you estimate the standard error, i.e. $\text{SE}(\hat{\theta}_1)$.

Solution

- (a) We have that $E[\hat{\theta}_1] = E[\hat{\beta}_1 + \hat{\beta}_2] = E[\hat{\beta}_1] + E[\hat{\beta}_2] = \beta_1 + \beta_2 = \theta_1$, where the penultimate equality comes from the unbiasedness of the OLS estimator and the last equality is by the definition of θ_1 .

(b)

$$\begin{aligned} \text{Var}(\hat{\theta}_1) &= \text{Var}(\hat{\beta}_1 + \hat{\beta}_2) = \text{Var}(\hat{\beta}_1) + \text{Var}(\hat{\beta}_2) + 2\text{Cov}(\hat{\beta}_1, \hat{\beta}_2) \\ &= \text{Var}(\hat{\beta}_1) + \text{Var}(\hat{\beta}_2) + 2\text{Cov}(\hat{\beta}_1, \hat{\beta}_2) \end{aligned}$$

- (c) There are two ways to estimate the standard error.

- Obtain an estimate of the theoretical variance, say $\widehat{\text{Var}}(\hat{\theta}_1)$ and then compute $\text{SE} = \sqrt{\widehat{\text{Var}}(\hat{\theta}_1)}$. To obtain $\widehat{\text{Var}}(\hat{\theta}_1)$ notice that each component of the theoretical variance $\text{Var}(\hat{\theta}_1) = \text{Var}(\hat{\beta}_1) + \text{Var}(\hat{\beta}_2) + 2\text{Cov}(\hat{\beta}_1, \hat{\beta}_2)$ can be estimated with $\widehat{\text{Var}}(\hat{\beta}_1)$, $\widehat{\text{Var}}(\hat{\beta}_2)$ and $\widehat{\text{Cov}}(\hat{\beta}_1, \hat{\beta}_2)$, respectively. These estimates can be obtained from the estimated variance-covariance matrix when we estimate the model $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + u$. Thus, we get that

$$\widehat{\text{Var}}(\hat{\theta}_1) = \widehat{\text{Var}}(\hat{\beta}_1) + \widehat{\text{Var}}(\hat{\beta}_2) + 2\widehat{\text{Cov}}(\hat{\beta}_1, \hat{\beta}_2).$$

Then, we take the square root of the previous expression to get the standard error:

$$\begin{aligned} \text{SE}(\hat{\theta}_1) &= \sqrt{\widehat{\text{Var}}(\hat{\beta}_1) + \widehat{\text{Var}}(\hat{\beta}_2) + 2\widehat{\text{Cov}}(\hat{\beta}_1, \hat{\beta}_2)}, \\ &= \sqrt{\text{SE}(\hat{\beta}_1)^2 + \text{SE}(\hat{\beta}_2)^2 + 2\widehat{\text{Cov}}(\hat{\beta}_1, \hat{\beta}_2)}. \end{aligned}$$

Remark: $\text{SE}(\hat{\theta}_1) \neq \text{SE}(\hat{\beta}_1) + \text{SE}(\hat{\beta}_2) + 2\sqrt{\widehat{\text{Cov}}(\hat{\beta}_1, \hat{\beta}_2)}$.

- Transform the regression equation so that you can get the standard error directly. We will add and subtract the term $\beta_2 x_1$ so that the term $\theta_1 = \beta_1 + \beta_2$ appears in the equation:

$$\begin{aligned} y &= \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + u, \\ &= \beta_0 + \beta_1 x_1 + \beta_2 x_1 - \beta_2 x_1 + \beta_2 x_2 + \beta_3 x_3 + u, \\ &= \beta_0 + (\beta_1 + \beta_2)x_1 + \beta_2(x_2 - x_1) + \beta_3 x_3 + u, \\ &= \beta_0 + \theta_1 x_1 + \beta_2 R + \beta_3 x_3 + u, \end{aligned}$$

where $R = x_2 - x_1$. Estimating this new equation will yield an estimate $\hat{\theta}_1$ which consistently estimates θ_1 . Furthermore, we will be able to get the standard error $SE(\hat{\theta}_1)$ from the output of this regression.

5. Consider the multiple regression model with three independent variables, under the four OLS assumptions.

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + u.$$

You would like to test the null hypothesis $H_0 : \beta_1 - 3\beta_2 = 1$.

- (a) You can follow the following procedure: Let $\hat{\beta}_1$ and $\hat{\beta}_2$ denote the OLS estimators of β_1 and β_2 . Find $\text{Var}(\hat{\beta}_1 - 3\hat{\beta}_2)$ in terms of the variances of $\hat{\beta}_1$ and $\hat{\beta}_2$ and the covariance between them. What is the standard error of $\hat{\beta}_1 - 3\hat{\beta}_2$? Now, using the standard error that you found, write the t -statistic for testing $H_0 : \beta_1 - 3\beta_2 = 1$.
- (b) Define $\theta_1 = \beta_1 - 3\beta_2$ and $\hat{\theta}_1 = \hat{\beta}_1 - 3\hat{\beta}_2$. Write a regression equation involving $\beta_0, \theta_1, \beta_2$ and β_3 that allows you to directly obtain $\hat{\theta}_1$ and its standard error.

Solution

- (a) We have $\text{Var}(\hat{\beta}_1 - 3\hat{\beta}_2) = \text{Var}(\hat{\beta}_1) + 9\text{Var}(\hat{\beta}_2) - 6\text{Cov}(\hat{\beta}_1, \hat{\beta}_2)$. This is the true variance of the expression. To get an estimate of this variance we compute $\widehat{\text{Var}}(\hat{\beta}_1 - 3\hat{\beta}_2)$ which is equal to $\widehat{\text{Var}}(\hat{\beta}_1) + 9\widehat{\text{Var}}(\hat{\beta}_2) - 6\widehat{\text{Cov}}(\hat{\beta}_1, \hat{\beta}_2)$. Thus the standard error of $\hat{\beta}_1 - 3\hat{\beta}_2$ is just the square root of the previous estimate:

$$\sqrt{\widehat{\text{Var}}(\hat{\beta}_1) + 9\widehat{\text{Var}}(\hat{\beta}_2) - 6\widehat{\text{Cov}}(\hat{\beta}_1, \hat{\beta}_2)} = \sqrt{SE(\hat{\beta}_1)^2 + 9SE(\hat{\beta}_2)^2 - 6\widehat{\text{Cov}}(\hat{\beta}_1, \hat{\beta}_2)}.$$

Thus, the t -statistic is:

$$t = \frac{\hat{\beta}_1 - 3\hat{\beta}_2 - 1}{\sqrt{SE(\hat{\beta}_1)^2 + 9SE(\hat{\beta}_2)^2 - 6\widehat{\text{Cov}}(\hat{\beta}_1, \hat{\beta}_2)}}$$

(b)

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + u.$$

$$y = \beta_0 + \beta_1 x - 3\beta_2 x_1 + 3\beta_2 x_1 + \beta_2 x_2 + \beta_3 x_3 + u$$

$$y = \beta_0 + (\beta_1 - 3\beta_2)x_1 + \beta_2(3x_1 + x_2) + \beta_3 x_3 + u$$

$$y = \beta_0 + \theta_1 x_1 + \beta_2 R + \beta_3 x_3 + u$$

where $R = 3x_1 + x_2$. Estimating this regression, allows us to obtain the estimate $\hat{\theta}_1$ and its standard error $SE(\hat{\theta}_1)$, which is exactly what we obtained in part (a).

6. Consider the homoskedasticity only F -statistic

$$F = \frac{(R^2_{unrestricted} - R^2_{restricted})/q}{(1 - R^2_{unrestricted})/(n - k_{unrestricted} - 1)},$$

and show that it can also be written as

$$F = \frac{(SSR_{restricted} - SSR_{unrestricted})/q}{SSR_{unrestricted}/(n - k_{unrestricted} - 1)}$$

Solution

Notice that $1 - R^2_{unrestricted} = 1 - (1 - SSR_{unrestricted}/TSS) = SSR_{unrestricted}/TSS$, and that $R^2_{restricted} - R^2_{unrestricted} = (1 - SSR_{unrestricted}/TSS) - (SSR_{restricted}/TSS) = (SSR_{restricted} - SSR_{unrestricted})/TSS$, which yields the result.

7. Empirical exercise. Multiple Linear Regression.

For this exercise, use the *Birthweight_Smoking* data set posted on BlackBoard, which contains data for a random sample of babies born in Pennsylvania in 1989. The data include the baby's birth weight together with various characteristics of the mother, including whether she smoked during the pregnancy. You can find a detailed description of the dataset in the file *Birthweight_Smoking_Description*. Use this dataset to answer the following questions.

- (a) Regress *Birthweight* on *Smoker*. What is the estimated effect of smoking on birth weight?
- (b) Regress *Birthweight* on *Smoker*, *Alcohol*, and *Nprevist*.
 - i. Explain why the exclusion of *Alcohol* and *Nprevist* could lead to omitted variable bias in the regression estimated in (a).
 - ii. Is the estimated effect of smoking on birth weight substantially different from the regression that excludes *Alcohol* and *Nprevist*? Does the regression in (a) seem to suffer from omitted variable bias?
 - iii. Jane smoked during her pregnancy, did not drink alcohol, and had 8 prenatal care visits. use the regression to predict the birth weight of Jane's child.
 - iv. Compute R and $\bar{R}^2$. Why are they so similar?
- (c) An alternative way to control for prenatal visits is to use the binary variables *Tripre0*, through *Tripre3*. Regress *Birthweight* on *Smoker*, *Alcohol*, *Tripre0*, *Tripre2* and *Tripre3*.
 - i. Why is *Tripre1* excluded from the regression? What would happen if you included it in the regression?
 - ii. The estimated coefficient on *Tripre0* is large and negative. What does this coefficient measure? Interpret its value.
 - iii. Interpret the value of the estimated coefficients on *Tripre2* and *Tripre3*.
 - iv. Does the regression in (c) explain a larger fraction of the variance in birth weight than the regression in (b)?

8. Empirical exercise. Use the data in HPRICE1 for this exercise.

- (a) Estimate the model

$$price = \beta_0 + \beta_1 lotsize + \beta_2 sqrft + \beta_3 bdrms + \beta_4 colonial + u,$$

where *lotsize* is size of lot in square feet, *sqrft* is size of house in square feet, *bdrms* is number of bedrooms and *colonial* is a dummy variable that equals 1 if the home is colonial style. Report the results of your estimations in the usual form, including the standard error of the regression.

- (b) Obtain a predicted price, when we plug in $lotsize = 10000$, $sqrft = 2300$, $bdrms = 4$ and $colonial = 1$; round this price to the nearest dollar.
- (c) Run a regression that allows you to put a 95% confidence interval around the predicted value in (b).

- (d) Suppose that house A has characteristics as in (b). Now consider another house, B which has $lotsize = 10000$, $sqrft = 2400$, $bdrms = 5$ and $colonial = 0$. Let $\hat{P}_A$ and $\hat{P}_B$ be the predicted price of A and B respectively. Let $\widehat{\Delta P} = \hat{P}_A - \hat{P}_B$ denote the expected change in price. Compute a 95% confidence interval for this change in price. Do this by transforming the regression, similarly as you did in items 4–5, to get the standard error.